Md. Nazmul (Corresponding Author)
Department of Computer Science and Engineering
College of Technology, National University
Gazipur 1704, Bangladesh
Email: nazmulhas36@gmail.com
ORCID: 0009-0001-6115-7023

Date: November 1, 2026

To:
The Editor-in-Chief
Engineering Applications of Artificial Intelligence (EAAI)
Elsevier Science

Subject: Submission of Original Research Manuscript for Double-Blind Peer Review

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript entitled:

"Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage"

for consideration as a regular research paper in *Engineering Applications of Artificial Intelligence*.

Scope and Alignment with EAAI Mandate:
In direct alignment with the core mandate of *Engineering Applications of Artificial Intelligence*, this work develops, implements, and empirically validates an applied, industrial-scale artificial intelligence system—named **C-STGB** (Conformal Spatio-Temporal GraphBoost)—engineered for real-time financial transaction surveillance and automated anti-money laundering (AML) compliance. We confirm that this work contains **no military or defense applications**. Addressing real-world engineering constraints in financial institutions, C-STGB provides sub-millisecond (0.45 ms) streaming inference, finite-sample statistical risk guarantees, and resilience against adversarial topology manipulation, overcoming five foundational bottlenecks: microsecond velocity evasion, topological camouflage, extreme class imbalance ($\pi < 0.05\%$), cold-start account sparsity, and distribution drift.

Key Engineering Contributions and Substantive Findings:

- **Continuous-Time Multi-Scale Temporal Graph Architecture:** C-STGB unifies continuous harmonic temporal embeddings, Hawkes arrival intensity, and a context-aware edge-trust filter ($\hat{g}_{ij}$) that prunes 65.9% of camouflage chaff connecting to high-degree banking hubs.
- **Flow-Conserving Typology Oversampling:** A latent GraphSMOTE mechanism clustered by laundering archetypes with Asymmetric Focal Tversky optimization, formally proven to preserve point-level mass-flow conservation invariants.
- **Finite-Sample Risk Guarantees under Drift:** Class-Conditional Conformal Risk Control establishing $\geq 99.0\%$ empirical true-label coverage under within-class exchangeability, non-asymptotic drift bounds, and online Martingale tracking via Adaptive Conformal Inference (ACI). Decisive singletons route $> 99.4\%$ of volume through automated straight-through tiers, reducing officer review queues to 0.50%.
- **Large-Scale Empirical Corpus & System Latency:** Evaluated across 14 financial transaction networks (9.53M entities and 9.53M transactions, totaling $9,534,980$ entities and $9,528,355$ transactions across 182 evaluation pairs over five locked seeds), C-STGB achieves 68.05% unweighted macro F1 (0.6974 PR-AUC), delivering a $+3.29$ pp uplift over native-feature XGBoost (64.76%, $p = 0.0032$) and $+44.27$ pp mean uplift over deep spatial GNNs ($W = 105.0, p_{\text{adj}} < 0.001$). Streaming Fast-Path inference executes in 0.45 ms per single event ($31,200$ tx/s), fully compliant with Tier-1 banking latency SLAs.

- **Compliance Decision Support Prototype:** An auxiliary module assisting examiners in compiling verifiable FinCEN Form 111 XML dossiers under Federal Reserve SR 26-2 / OCC Bulletin 2011-12 model risk governance standards.

Submission Declarations & Policy Compliance:

- **Double-Blind Peer Review:** In strict compliance with EAAI policy, two separate files are provided: (1) an unanonymized Title Page with all author details, affiliations, ORCIDs, CRediT statement, and acknowledgments; and (2) a completely blinded manuscript with zero identifying information and anonymized repository links.
- **Research Data Policy:** All 14 benchmark datasets are publicly accessible. Complete pipelines, configuration files, and reproduction scripts have been deposited in an anonymized repository (https://anonymous.4open.science/r/cstgb-anonymous-audit) and will be archived with a permanent Zenodo DOI upon publication.
- **Declaration of Generative AI:** Disclosed in a dedicated mandatory section preceding the references.
- **Institutional Thesis Origin Disclosure:** We disclose that an earlier exploratory version of this work was developed as an undergraduate B.Sc. thesis by the first author at the National University of Bangladesh. In accordance with Elsevier originality guidelines, the current submission to EAAI represents a substantially expanded, re-engineered scientific manuscript featuring advanced continuous-time tri-band attention, the formalization and non-asymptotic derivation of Class-Conditional Conformal Prediction under distribution shift, an expanded 14-network empirical benchmark corpus, comprehensive non-parametric statistical testing, and an industrial-grade latency profiling study.
- **Originality:** This manuscript is original, has not been published previously, and is not under review elsewhere.

Thank you for your consideration of our manuscript. We look forward to receiving the reviewers' constructive feedback.

Sincerely yours,

Md. Nazmul (Corresponding Author, nazmulhas36@gmail.com)
Musrat Jahan Gungun (gungunjahan84@gmail.com)
Maheli Ahmed (Faculty Supervisor, maheli.ahmed.cse.cot@gmail.com)
Department of Computer Science and Engineering
College of Technology, National University, Gazipur 1704, Bangladesh